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An extended logarithmic visualization improves forecasting accuracy for exponentially growing numbers, but residual difficulties remain☆

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ABSTRACT

Humans find it notoriously difficult to predict the future development of numbers in scenarios where the data exhibits exponential growth. This study explored how employing logarithmically scaled graphs can improve forecasting accuracy in such scenarios. Experiment 1 shows that a modified visualization improves forecasting, mitigating the inaccuracies encountered with linear and ordinary logarithmic depictions. The modification consists of putting the y-axis on the right side of a logarithmically scaled graph and extending the x-axis to the range of the forecast period. This effect was independent of general graph literacy, and participants were more confident in their estimates. To uncover the role of tick marks in estimation accuracy, we conducted a second experiment manipulating the presence of minor tick marks and varying target values systematically with respect to their proximity to the next major tick mark. Participants performed worse for target values midway between two major tick marks and no accuracy benefits related to the presence of tick marks. Analysis of eye movements during the same task suggests that the poor utilization of minor tick marks is not simply due to a lack of attention but to difficulties in converting the location into the corresponding numerical value.

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1. Introduction

The COVID-19 pandemic brought (renewed) awareness to the fact that humans are poor at predicting future numbers if the respective growth is exponential. The systematic underestimation in forecasting exponentially growing numbers has been termed *exponential growth bias* (EGB, Wagenaar & Timmers, 1979). In graphs plotting cases against days, exponential growth is challenging to spot in its early phase if the scale of the visualization is linear (e.g., Perneger, Kevorkian, Grenet, Gallée, & Gayet-Ageron, 2020). Fig. 1 exemplifies these problems. In the present study, we evaluated different features of graphs depicting exponential growth.

During the COVID-19 pandemic, a realistic, unbiased apprehension of exponential growth would have been a valuable skill (e.g., Banerjee, Bhattacharya, & Majumdar, 2021). With emerging infectious diseases on the rise (Jones et al., 2008), this utility is not waning. Educative means employed to mitigate EGB - including educating participants about the nature of exponential growth (Wagenaar & Sagaria, 1975); but see (Hamann, 2022) and making them aware of the bias (Schonger & Sele, 2020) - have failed. Other methods have proven useful, such as asking participants to point to instead of providing a number (Ciccione, Sablé-Meyer, & Dehaene, 2022) and using doubling rates rather than growth rates to communicate

☆ The numerical results presented in this manuscript were reproduced (up to some minor discrepancies) on the 20th of September 2024.

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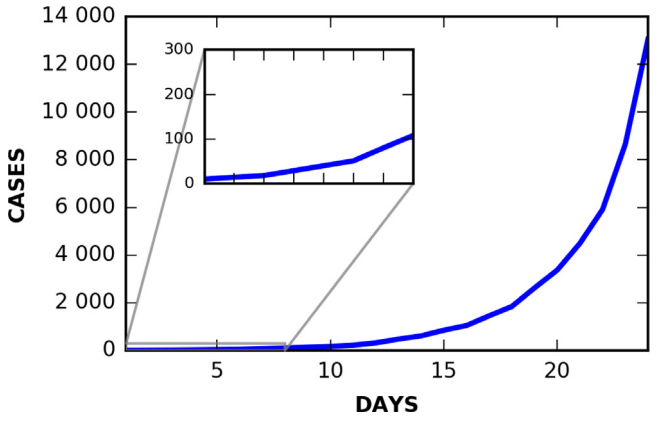


Fig. 1. Cases by days depicted in an ordinary linear plot. Exponential growth is challenging to detect in the early phase of growth, and concrete future values are very hard to forecast from visual inspection alone when the scale of the visualization is linear.

growth (Schonger & Sele, 2020). Lammers, Crusius, and Gast (2020) found increased support for measures such as social distancing and lockdowns in real-world judgments when doubling rates were used.

When it comes to graphical depictions of exponential growth, an important means to mitigate EGB would be to aid participants' graph comprehension. An enhanced understanding of a graph's information directly translates to attenuating the EGB. For instance, using a logarithmically scaled *y*-axis substantially reduces the EGB (Hutzler et al., 2021). In that study, participants were presented with illustrations of exponential growth (with different growth rates) for the initial 20 days of a hypothetical epidemic, either on a linear or logarithmic scale. Their task was to estimate the number of new infections on day 30. Participants underestimated the target values substantially when the data were presented on a linear scale. When the graphs were logarithmically scaled, their estimates were closer to the target, but there still was a substantial bias to underestimate the final number. As using different scales in the depiction of health related data affects people's policy preferences (e.g., Romano et al., 2020), it is worthwhile to further investigate this effect.

In the present study, we examine the effects of specific features of exponential data and its presentation on people's ability to apprehend its progression. We made three adjustments to the features of graphs meant to convey this information, the latter two of which have hitherto not been examined:

1. The scale of the graph is logarithmic.
2. The *y*-axis is on the right side and its limit contains the to-be-estimated target value.
3. The *x*-axis is extended to the date the estimate should be made, showing the whole range of the forecast period.

An example is provided in the right panel of Fig. 2. We refer to the type of depiction as *log-extended*. In the first experiment, we assessed participants' ability to forecast exponential growth using this modified layout compared to the linear and ordinary logarithmic layout.

In a second experiment, we further varied features of the presentation of exponentially growing data. Two recent studies investigating EGB for line graphs and scatterplots found that logarithmic scaling of the *y*-axis led to an overestimation of exponential growth (Ciccione et al., 2022; Melnik-Leroy et al., 2023). Two related explanations have been provided for this particular bias. Firstly, it was suggested that although logarithmic scaling generally facilitates the mere estimation of future developments, it impedes a thorough understanding of the given data. Linear scaling, on the other hand, enables an intuitive understanding of the data but makes predicting future developments more challenging (Melnik-Leroy et al., 2023). This explanation suggests that a drawback of logarithmic scaling might lie in a tendency to overestimate, implying equally detrimental downsides to linear and logarithmic scaling.

Intending to enhance estimation accuracy, we presume a clear superiority of logarithmic scaling over linear scaling and sought to understand further mechanisms involved in a potential overestimation. To further understand the features of the graphs that might have contributed to the overestimation, we thus turn to the more technical explanation provided by Ciccione et al. (2022). The authors suggested that participants had difficulties relating a given point on a logarithmically scaled axis to a corresponding numerical value because the dispersion of values along a logarithmically scaled axis is counterintuitive. Participants might have linearly interpolated between major tick marks to provide a numerical estimate. The value corresponding to the point halfway between 1000 and 10,000 might be guessed to be 5000 when it is 3160. Participants' difficulties might lie in converting a located point on the axis into the correct numerical value rather than locating it correctly in the first place. This implies that participants may be more accurate in estimating numbers whose corresponding minor tick mark is close to a major tick mark (e.g., 2000 and 8000) and less accurate for values whose corresponding minor tick mark is further away from the major tick marks.

In the second experiment, we tested this explanation and the involved cognitive mechanism(s) for the reversed estimation tendency. To do so, we compared participants' estimates for log-extended graphs with and without minor tick marks for a specific set of target values (see

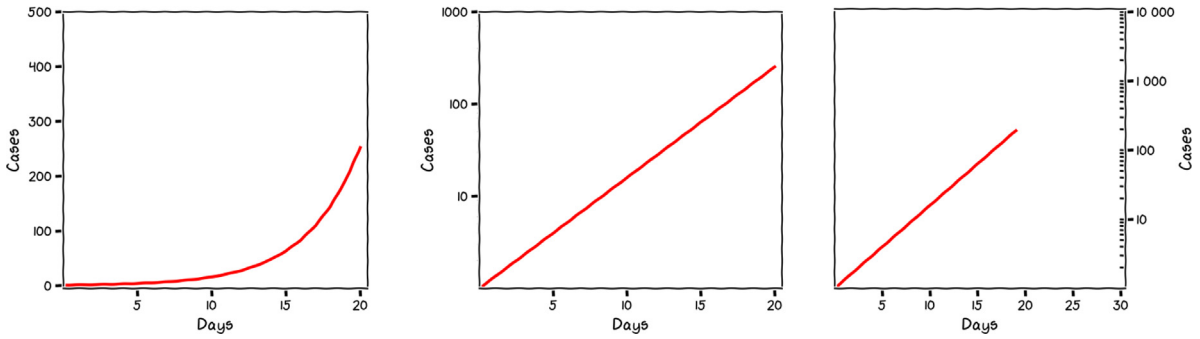


Fig. 2. Example of the stimuli. The left graph shows the data in an ordinary linear scale. The graph in the middle is on an ordinary logarithmic scale. The graph to the right is also log-scaled but additionally has an extended *x*-axis, and the *y*-axis is on the right side. This may allow a better projection of the future number of infections. In each of these examples, the number of infections on day 30 is 4000 (growth rate: 1.31847).

below). We repeated the same task in a third experiment and recorded participants' eye movements. By investigating their gaze behavior, we sought to assess whether participants took notice of the minor tick marks and thus - provided they do understand their properties - were at least given the opportunity to improve their estimations.

2. Experiment 1

The general experimental framework of the first experiment was adopted from Hutzler et al. (2021): Graphs represented the progression of the number of infections over the first 20 days of a hypothetical epidemic, and participants had to estimate the number of cases on day 30. Those graphs were presented using linear, logarithmic, and log-extended scales. A subsample of the participants was also asked about their confidence in their estimate. In addition, we assessed participants' *graph literacy* (GL), that is, their ability to understand graphically presented (health-related) data, to examine a potential interdependency between this (more) general skill and the correct apprehension of exponential growth.

Hypotheses. We expected a significant effect of scale on estimation accuracy. According to previous evidence (Hutzler et al., 2021), participants should be more accurate in predicting the trajectory of exponential data in the log-extended condition than in the ordinary logarithmic condition and more accurate in the ordinary logarithmic than in the linear condition. Furthermore, we expected estimates from participants with higher general graph literacy to be more accurate than those from participants with lower general graph literacy.

2.1. Methods

Participants and data preprocessing. Participants were recruited via the platform Sona Systems and received course credit and the chance to win a €50 voucher for an online bookstore. This remuneration was independent of their task performance, meaning there was no monetary incentive for providing more accurate estimations. We collected the data from two independent samples. The size of Sample 1 was 92 participants, and the size of Sample 2 was 136 participants (who completed the survey). The two samples viewed plots depicting exponentially growing data (20 days), and their task was to predict the number on day 30 (see below; for the exact instructions, see Supplementary Material S6). For the analysis, the participants' estimates were converted into percentages of the target value, as the absolute values to be estimated differed between the two samples. We conducted permutation tests assessing the exchangeability of the two datasets. As they revealed no differences with respect to central tendencies and confidence intervals (see Supplementary Material S1), we combined the two datasets into one. Thus, we had a sample size of 228 participants (147 female, 80 male, and one diverse). All participants were native speakers of German, and their ages ranged from 15 to 80 years, with a mean of 28.41 years, a median (*Md*) of 22.0 years, and an interquartile range (*IQR*) of 20.0 to 26.0 years.

Procedure. In an online survey application, participants were presented with plots with linear, common logarithmic, and log-extended scales (Graph Condition), as shown in Fig. 2. All plots depicted the first 20 days of the growth of a hypothetical epidemic (in terms of daily new cases). The participant's task was to predict the number of new cases on day 30 by typing in the estimated value. Sample 1 saw graphs with target values of 2000, 6000, and 10,000 and hence a total of nine different graphs (3 scales *x* 3 exponential growth functions). Sample 2 saw graphs with target values of 4000 to 9000 with increments of 1000, which amounts to six different target values and a total of 18 graphs. The *y*-axis ranges were identical within graph conditions (see Fig. 2). Each participant's order of target values was randomized within the same graph conditions. Sample 2 additionally rated the confidence in their estimates ("*How confident were you about the last three estimations?*") on a scale from 0 to 100. The two surveys differed in other minor aspects irrelevant to the present study. Both samples completed a test of *graph literacy*, that is, the ability to interpret graphically depicted (health-related) data.

Graph literacy. For measuring graph literacy (GL), we used the Short Graph Literacy Scale (SGLS) by Okan, Janssen, Galesic, and Waters (2019), which we translated into German. The test is a short version of a larger scale with only four items (Galesic & Garcia-Retamero, 2011). According

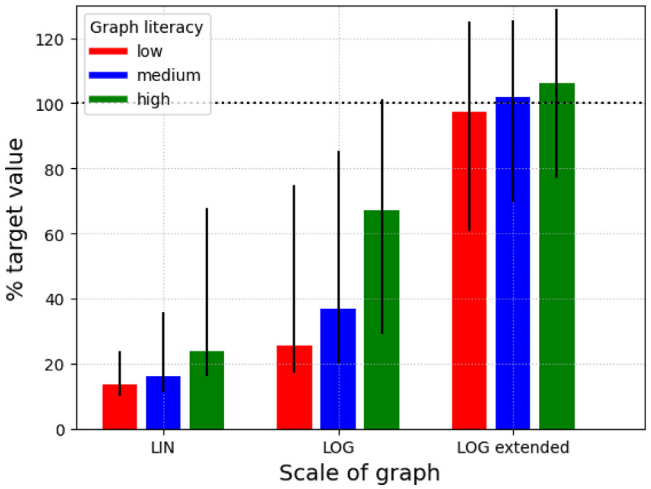


Fig. 3. Medians of the estimated percentage of the target values for the different scales separately for the three groups of graph literacy (low, medium, and high as measured using the Short Graph Literacy Scale by Okan et al., 2019). Bars show the IQR.

to the authors, the high discriminant and predictive validity of the SGLS makes it preferable to the more extensive version, as it is more time-efficient and, hence, less demanding for participants. Participants were divided into three groups: low ($N = 83$), medium ($N = 89$), and high ($N = 56$) GL. Participants with low GL answered fewer than three items correctly, participants with medium GL answered three items correctly, and participants with high GL answered all four items correctly.

Apparatus. The online surveys were conducted using *LimeSurvey* (version 3.22.2+200204, Hamburg, Germany). Data analyses were conducted using Python as the programming language.

2.2. Results and discussion

Due to considerable variance in the data, we report the median and interquartile range (IQR, i.e., percentile 25 and 75 of the data). We estimate the median's 95% confidence interval (CI) to estimate statistically significant differences between the conditions. The CIs were estimated as the 2.5 percentile and 97.5 percentile of 10,000 random permutations of the data. Differences between conditions (scale of the graphs) and groups (graph literacy) are considered statistically significant if the respective CIs do not overlap.

Fig. 3 summarizes participants' estimates expressed in percentage of the target value. Participants considerably underestimated the target value of the linear graphs, with a median estimate of only 16.90% of the actual target value [CI: 14.70–19.33]. Participants gave significantly higher estimates for the graphs with the ordinary logarithmic scale, *Md* = 35.56% [CI: 30.84–42.42], but underestimated the actual target value. Note that these median estimates significantly differ from the target value (i.e., 100%) as the CIs did not include it. The median estimates for the graphs with the extended *log*-scale are significantly higher than the former estimates (*Md* = 102.74%) and not significantly different from the actual target value, as the CI of 96.79 to 108.33 includes the target value.

With regard to the effect of the participants' general graph literacy (GL) on the estimation of the target values, Fig. 3 shows a (somewhat) better performance of the group with high GL compared to the group with low GL. This effect seemed to be most pronounced for ordinary log-scale graphs and least for graphs with the extended *log*-scale. The analysis confirmed this pattern. For the graphs with the ordinary log-scale, the estimates of the high GL group were significantly higher (i.e., closer to the true value) than those of the low GL group [*Md* = 67.11; CI: 36.78–89.14 vs. 25.56; 20.39–31.67]. This comparison was also significant for the graphs with linear scale [*Md* = 23.99; CI: 20.00–40.70 vs. 13.70; 12.32–16.11]. The group difference was not significant for the graphs with the extended log-scale [*Md* = 106.06; CI: 91.32–116.67 vs. *Md* = 97.26; CI: 85.09–111.11]. The differences in the median estimates of the low and high GL groups and those of the medium GL group were not significant.

The analysis of confidence ratings (in Sample 2) revealed the lowest confidence for the ordinary log-scale [*Md* = 40; CI: 35–45] and the highest (and significantly higher) confidence for the extended log-scale [*Md* = 65; CI: 55–70]. The median confidence of the estimates for linearly scaled plots was 50 [CI: 40–55]. The reasons for higher confidence for the log-extend graphs were presumably twofold: First, the extended *x*-axis - covering the entire forecasting period - allowed participants to extend the trajectory to the point on the *y*-axis on day 30. Second, the point on the *y*-axis could be pinpointed directly as the *y*-axis was already positioned on the right-hand side. Both factors likely contributed to an impression of straightforwardness of the task in the log-extended condition. The linear and ordinary log-scale likely led to lower confidence because participants had to continue the log-scaled *y*-axis mentally. In the linear condition, however, participants were at least intuitively familiar with linearly depicted data, resulting in higher confidence ratings.

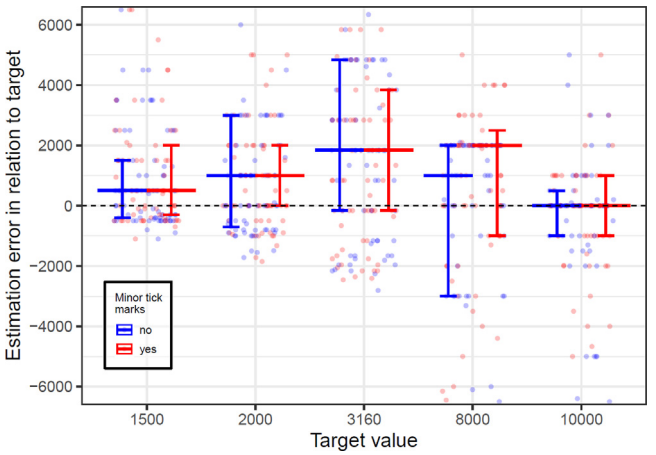


Fig. 4. Median estimation errors by target values for the two tick mark conditions. The dashed line represents estimation error = 0. Error bars indicate the IQR. The markers are individual estimates. The *y*-axis is visually truncated; hence, the most extreme estimates are not displayed (11.4% and 11.7% of the data in the no-ticks and ticks condition, respectively).

3. Experiment 2

One feature of the log-extended graphs that may have contributed to the improvement in estimation accuracy found in Experiment 1, but has not been explicitly investigated, is the minor tick marks on the logarithmically scaled *y*-axis. As can be seen in Fig. 2, the log-extended graph uses a well-annotated *y*-axis with minor tick marks at an interval of 1/10 between the major tick marks. However, the representation of graphs in different everyday situations is not always sophisticated. In the second experiment, we relate the variation of this feature to the recent finding that logarithmic scaling can lead to an overestimation of exponential growth (Ciccione et al., 2022), independent of participants' mathematical education (Melnik-Leroy et al., 2023). While an overestimation seems preferable to an underestimation of exponential growth in some practical scenarios, it can still be a dangerous bias. Here, we examined (i.) the effect of the presence vs. absence of minor tick marks and (ii.) the role of the distance of the target values from the major tick marks.

Hypotheses. Our expectations were twofold: First, we expected a significant improvement in estimation accuracy for conditions with minor tick marks (compared to those without minor tick marks). Second, we expected a significant effect of target value on estimation accuracy: Generally, the further away the target values are from the major tick marks, the less accurate the estimate might be. The estimation error should be largest for target values midway between two major tick marks.

3.1. Methods

Data preprocessing. Eighty-four out of 107 participants completed the survey. Five participants were excluded because they took more than 60 s to respond to stimuli several times. Two participants were excluded because they reported not seeing the presented graphs properly. This left a total number of 77 participants (46 female, 26 male, five diverse). All participants were native speakers of German, and their ages ranged from 18 to 61 years, with a mean of 23.47 years, a median (Md) of 21.0 years, and an interquartile range (IQR) of 20.0 to 23.0 years.

Procedure. The second experiment specifically tested the explanation for the abovementioned logarithmic scale bias. It results from difficulties converting a point on a logarithmically scaled axis into a numerical value (Ciccione et al., 2022). For this, we again used the scenario described above and the same online setup (the instructions were also highly similar; see Supplementary Material S6). This time, however, we used only log-extended graphs with the additional manipulation of inserting or omitting the minor tick marks on the *y*-axis for a different set of target values. Target values were chosen to optimally display the conditions described in our hypothesis, namely 1500, 2000, 3160, 8000, and 10,000. The *y*-axis range was the same for all graphs (0–10,000). The ten graphs were presented in two blocks – one with the graphs without minor tick marks and one with the graphs with minor tick marks. The graphs within the blocks were randomized. The reason for dividing the stimuli into two blocks and always presenting the no-tick mark block first was to prevent a learning effect. Once participants have been confronted with graphs with minor tick marks, they would most likely understand the property of a logarithmically scaled axis. Though an interesting question to investigate, the learning effect due to mere exposure was not in the scope of this experiment. Graph literacy was again assessed with the Short Graph Literacy Scale (Okan et al., 2019).

Apparatus. As for Exp. 1, the online surveys were done with *LimeSurvey* (version 3.26.1+210427, Hamburg, Germany).

3.2. Results and discussion

This time we report the estimation error (rather than the percentage of the target value due to its target-value

dependence). The median estimation errors are presented - separately for the different target values - in Fig. 4. As in Experiment 1, confidence intervals (CI) were estimated as the 2.5 percentile and 97.5 percentile of 10,000 random permutations of the data. A median estimation error (or the difference between two) is considered statistically significant if the respective CI does not include 0 (or does not overlap). All medians and estimated CIs can be viewed in a table in Supplementary Material S3.

Generally, the median estimate (*Md* = 112.5%) was slightly higher than in Exp. 1, confirming the expected overestimation for this set of target values. In particular, participants significantly overestimated the target values of 3160 and 8000. The median estimation error for the target value 3160 was 1840 for the no-tick and tick mark conditions [CI: 1840–2840 and 840–2840, respectively]. For the target value of 8000, the median errors were 1000 and 2000 [CI: 200–2000 and 400–2000]. Participants tended to overestimate the target values of 1500 and 2000, but this effect was not significant (except for the target value of 1500 in graphs with minor ticks). As a group, participants estimated the target value of 10,000 accurately with a median estimation error of 0, with 27 and 24 (out of 77) participants providing the exact value for the no-tick and tick condition, respectively. Another 11 and 20 were off-target by exactly ±1000. Thus, about half of the sample provided quite accurate estimates in this particular condition.

Comparisons of the estimates between the target values revealed that the error was significantly higher for the target value of 3160 than for the target values of 1500, 2000, and 10,000 in the no-tick mark condition. The difference was only significant in the tick mark condition compared to the target value of 10,000. Thus, the data partially confirmed our hypothesis that the target value of 3160 will lead to the highest estimation errors.

Suppose the estimation error for the target value of 3,160 - the point exactly in the middle between the major ticks marks of 1000 and 10,000 - was due to a "linearization" of the *y*-axis. In that case, 5000 should be a more common estimate than other mis-estimations. Indeed, 5000 was the most frequent estimate in the no-tick mark condition. However, this was not the case for the graphs with tick marks. Here, the most frequent estimates were 3000 and 6000. The estimates of 5000 and 4000 shared 3rd place in the most common answers.

Regarding the absence versus presence of minor tick marks, estimation errors did not differ. None of the estimates in the tick mark condition differed significantly from the estimates of the corresponding value in the no tick mark condition. Regarding graph literacy (GL), we replicated the finding of Exp. 1 that participants with different levels of GL literacy do not differ significantly from each other (for a graphical depiction, see Supplementary Material S2).

4. Experiment 3

In Experiment 2, minor ticks did not affect estimation accuracy. In Experiment 3, we investigated whether the participants attended to minor tick marks. To this end, we recorded their eye movements while they inspected the same graphs as in Experiment 2. As this study was conducted in the laboratory, the experimenter instructed participants verbally. Likewise, rather than typing their estimates, participants provided them verbally. Forty participants volunteered, none of whom also participated in Experiment 1 or 2. Three participants had to be excluded due to technical difficulties. This left 37 participants (26 female) aged 19 to 27 ($M = 21.5$; $Md = 21$).

Apparatus. The graphs were presented on a 16:9 monitor with a resolution of 1920 × 1080. Eye movements were recorded using an EyeLink 1000 (SR Research) desktop-mount system at a sampling rate of 1 kHz. Viewing was binocular, but only the right eye movements were recorded. The viewing distance, kept constant by a chin and forehead rest, was 60 cm.

4.1. Results and discussion

The participants of Experiment 3 were more accurate than the participants of Experiment 2 (and hence more similar to the participants of Exp. 1). We observed a significant misestimation for the target value 1500 and 8000 for the graphs without minor tick marks with a median estimate of 1000 [CI: 500–1300] and 9500 [CI: 8500–10,000]. This was the case for the graphs with minor tick marks for the target values of 1500 and 3160 with median estimates of 1000 [CI: 980–1000] and 2125 [CI: 2000–3000], respectively. All other median estimates did not differ significantly from the target value (for a graphical depiction, see Supplementary Material S4; for a table displaying all medians and estimated CIs, see S5).

Gaze behavior was used as a proxy for allocation of attention (Buswell, 1935). The heatmaps in Fig. 5 illustrate exemplarily the distribution of visual attention for graphs without and with minor tick marks. To analyze possible differences in the allocation of attention in the presence compared to the absence of minor tick marks, we defined an area of interest around the *y*-axis (±100 pixel in the horizontal dimension). In terms of the number of fixations, we found no significant difference with the presence of tick marks ($t(36) = 1.09$, $p = .282$), with M = 17.83 and 16.36 fixations in the vicinity of the *y*-axis for graphs without and with tick marks, respectively. Regarding the mean horizontal distance of the fixation location from the *y*-axis, there was a significant main effect with respect to tick marks, $t(36) = 2.32$, $p = .026$), with a mean distance of 0.74° visual angle for graphs with minor tick marks compared to 0.88° for graphs without minor tick marks. This numerically small (0.14°) but statistically significant effect suggests that participants' gaze was drawn more to the *y*-axis when it contained minor tick marks - corroborating the impression evoked by Fig. 5.

In a subsequent analysis, we assessed whether a closer inspection of the *y*-axis is associated with more accurate estimates. We computed Pearson correlations of the mean distance to the *y*-axis with the accuracy (operationalized as absolute error divided by target value). Neither of the two correlations was significant ($ps > .45$).

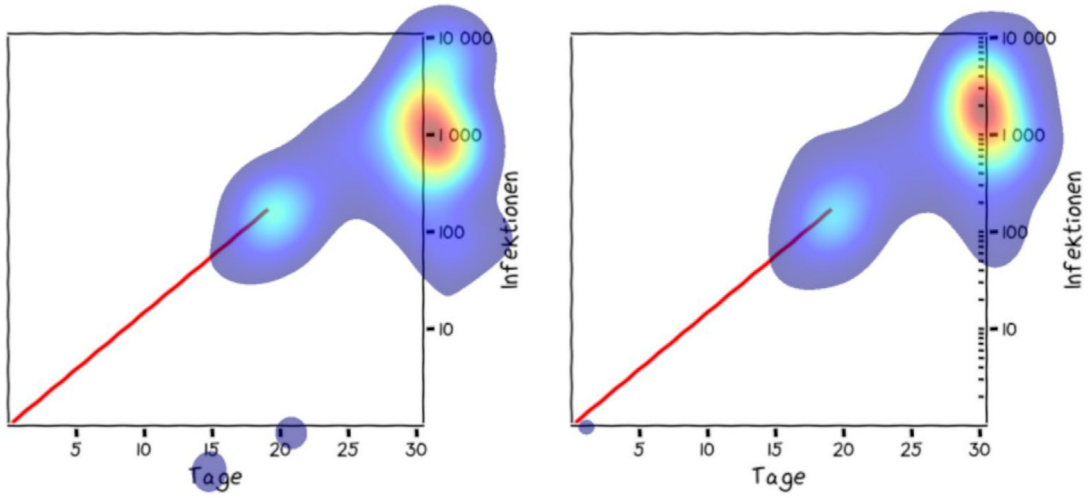


Fig. 5. Exemplary heatmap of the allocation of attention while viewing the graphs without and with minor tick marks. The target value in these graphs is 3160. This value has the farthest distance from major tick marks; it is in the exact middle between the major ticks of 1000 and 10,000. Heatmaps for the other stimuli (i.e., the different target values) can be found in the OSF repository.

5. General discussion

This study investigated the benefits and limitations of using logarithmically scaled graphs to improve estimation accuracy in simple graph comprehension tasks simulating situations of forecasting exponentially growing data. We investigated the effects of specific features on people's ability to anticipate exponential trajectories and make accurate predictions. In previous studies, log-scaled graphs improved the exponential growth bias (EGB) but did not eliminate it (e.g., Ciccione et al., 2022; Hutzler et al., 2021; but see Menge et al., 2018; Romano et al., 2020).

In a series of three experiments, we showed that prediction accuracy is improved when a so-called log-extended graph is used. Log-extended refers to the type of graph we used, in which the scales are logarithmic, the *y*-axis is on the right side and contains the value to be estimated, and the *x*-axis shows the whole range of the forecast period. Second, we could show that the accuracy of the forecast depends on the growth trajectory, which impedes the estimation if it happens to meet the y-axes about midway between major values (i.e., 10^1, 10^2, etc.). Third, our findings from Exp. 2, in combination with those of Experiment 3, indicate that the estimation errors in logarithmically scaled graphs are due to the conversion of visual estimates based on the logarithmic scale into actual numbers - rather than to problems of correctly locating the predicted outcome visually.

Evidence that the visual modifications to the ordinary logarithmic display led to more accurate estimates was provided by Experiment 1. As a result of extending the *x*-axis and placing the *y*-axis on the right-hand side, the anticipation of exponential trajectories was not only superior to the commonly used linear figures (replicating the findings of Hutzler et al., 2021; Melnik-Leroy et al., 2023) but also more accurate than ordinary logarithmic graphs. Significantly, participants benefited from the extended log-scale graphs (predicting more accurately and being more confident) regardless of their general graph literacy.

Given that some studies have found an overestimation of exponential growth when presented on logarithmic scales (Ciccione et al., 2022; Melnik-Leroy et al., 2023), Experiment 2 sought to understand what additional factors might increase prediction accuracy. Ciccione et al. (2022) proposed that participants have no difficulty visually identifying a target location on the *y*-axis for their estimates. They assumed that participants have difficulty assigning a numerical value to this location, that is, in converting the visual estimates into numerical values. We tested this assumption in Experiment 2. We showed that the increase in estimation accuracy when using log-extended graphs depends on the specific target values used. Minor ticks (thought to help participants with this conversion process) did not improve estimation accuracy. However, the spatial distance of the target values from the major tick marks mattered: Participants overestimated the target value to a greater extent than the other target values for the target value that lies midway between two major tick marks. Importantly, this finding suggests that the difficulty lies in the conversion. This conclusion is further supported by participants' gaze behavior analyzed in experiment 3. Their gaze indicated that they did mentally extend a given line toward the point on the *y*-axis and did take notice of minor tick marks. Thus, the crux probably lies in the next step, that is., the conversion into a numerical value. To further corroborate this, a future study could employ a task where participants simply click on a forecasted point on the *y*-axis (as was done by Ciccione et al., 2022) and additionally provide a numerical value.

The target value of 3160 in Experiment 2 is not only set apart from the others by its position lying midway between two major tick marks but also by the fact that it is not a round number (all the other target values in the Experiments were multitudes of 1000 or 500). Thus, one could also suggest that participants were biased to

provide round numbers. Our findings point toward a mixture of both: a linearization bias and a round number bias. The "linearized" value of 5000 was the participants' most common estimate for graphs without minor tick marks. For graphs with minor tick marks, 3000 (but also 4000, 5000, and 6000) were frequent estimates. The target value of 8000 was also prone to misestimations. This was presumably because this trajectory ended visually close to the major tick mark at 10,000. Future research should employ a greater variety of time series, including more diverse target values, to further investigate the mechanisms underlying these findings. The results from experiment 2 confirm the mitigation of the exponential growth bias by using log-extended graphs but cannot decisively explain the overestimation.

A surprising finding was that participants in Experiment 3 were generally more accurate than participants in Experiment 2. We can only speculate about potential reasons. A plausible explanation could be that the viewing conditions in the laboratory were quite optimal (large screen, good illumination). We could not control the viewing conditions in the online surveys. We have to assume that they varied - as testified by the occasional comments at the end of the survey that the graphs were hard to see. This could have been due to using devices that were too small (smartphones). Motivational factors might also play a role. Doing the estimations alone (online) is probably very different from doing them in the presence of another person (laboratory). However, although generally more accurate, the participants of Experiment 3 were not more accurate in the presence of minor tick marks than in the condition without minor tick marks.

Experiment 2 revealed that participants did not benefit from the presence of minor tick marks, and Experiment 3 revealed that this was not due to inattention or unawareness of this visual feature. Thus, as aforementioned, the problem seems to lie in converting the position of the tick mark to its corresponding numerical value (Ciccione et al., 2022). A logical fourth experiment would have been an educational intervention study with explicit tuition on converting the tick position into the corresponding value. Such an intervention study, however, has already been conducted. Melnik-Leroy et al. (2023) taught participants (who, in a former experiment, exhibited substantial misjudgments of exponential growth in logarithmically scaled graphs) how to read the minor tick marks correctly. Quite interestingly, the outcome of this intervention was mixed. Participants who already understood graphically depicted data (students of natural science/computer science) profited substantially from the intervention. In contrast, participants who were less familiar with graphs (students of foreign languages) profited only weakly. In the present study, participants benefited from the extended logarithmic visualization of exponentially growing numbers independent of their general graph literacy. That said, it is true that those participants with low graph literacy were most inaccurate in their estimates when the *y*-axis of the graph was on a linear or ordinary logarithmic scale. Thus, while explicit instruction on how to read logarithmically scaled axes seems - in general - helpful, future research should address the question of how instructions should be tailored for particular target populations.

Limitations. Two major limitations of this study need to be considered in order to evaluate its findings properly. This study investigated people's ability to forecast exponentially growing data visually (finding the correct spot on a graph) and numerically (converting this spot into a numerical value). However, all presented time series were entirely noise-free; that is, little to no uncertainty was involved in extrapolating the mere course of the numerical trend. Thus, of all the factors playing a role in the forecasting process, the findings of this paper should be understood as pertaining to graph comprehension. Furthermore, the set of target values (or growth rates) was not particularly diverse (mainly multitudes of 1000) and lacked ecological validity. They were specifically chosen to test our hypotheses experimentally, but their limited variety imposes constraints on generalizability.

6. Conclusion

Accurately forecasting numbers based on simple graphical depictions is a valuable skill for the public in general and policymakers in particular. In the event of an epidemic, for example, accurate judgments about the growth of the number of infections provide the basis for risk communication and well-informed countermeasures. Unfortunately, humans are inept at intuitively grasping exponential growth functions (e.g., Hutzler et al., 2021). Any means that assist in forecasting future values of exponentially growing numbers and graph comprehension are desirable. The present study showed that an extended logarithmic visualization improves forecasting accuracy, but also revealed that residual difficulties remain.

CRediT authorship contribution statement

Ben H. Engler: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Florian Hutzler:** Writing – review & editing, Supervision, Resources, Project administration. **Stefan Hawelka:** Writing – original draft, Visualization, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability statement

Stimuli, data and scripts of the analyzes will be publicly available upon acceptance of the manuscript. A preview of the project is available at: https://osf.io/3gj2k/?view_only=1ddb2bca605e4a02bce385a25ec30b70.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2024.09.006.

References

Banerjee, R., Bhattacharya, J., & Majumdar, P. (2021). Exponential-growth prediction bias and compliance with safety measures related to COVID-19. *Social Science & Medicine*, *268*(113473), http://dx.doi.org/10.1016/j.socscimed.2020.113473.

Buswell, G. T. (1935). *How people look at pictures: A study of the psychology and perception in art*. Chicago, IL: University of Chicago Press.

Ciccione, L., Sablé-Meyer, M., & Dehaene, S. (2022). Analyzing the misperception of exponential growth in graphs. *Cognition*, *225*, Article 105112. http://dx.doi.org/10.1016/j.cognition.2022.105112.

Galesic, M., & Garcia-Retamero, R. (2011). Graph literacy: A cross-cultural comparison. *Medical Decision Making*, *31*(3), 444–457. http://dx.doi.org/10.1177/0272989X10373805.

Hamann, H. (2022). On getting it right by being wrong: A case study of how flawed research may become self-fulfilling at last. *Proceedings of the National Academy of Sciences*, *119*(15), Article e2122274119. http://dx.doi.org/10.1073/pnas.2122274119.

Hutzler, F., Richlan, F., Leitner, M. C., Schuster, S., Braun, M., & Hawelka, S. (2021). Anticipating trajectories of exponential growth. *Royal Society Open Science*, *8*, Article 201574. http://dx.doi.org/10.1098/rsos.201574.

Jones, K. E., Patel, N. G., Levy, M. A., Storeygard, A., Balk, D., Gittleman, J. L., et al. (2008). Global trends in emerging infectious diseases. *Nature*, *451*(7181), 990–993. http://dx.doi.org/10.1038/nature06536.

Lammers, J., Crusius, J., & Gast, A. (2020). Correcting misperceptions of exponential coronavirus growth increases support for social distancing. *Proceedings of the National Academy of Sciences of the United States of America*, *117*(28), 16264–16266. http://dx.doi.org/10.1073/pnas.2006048117.

Melnik-Leroy, G. A., Aidokasm, L., Dzemyda, G., Dzemydaitè, G., Marcinkevicius, V., Tiesis, V., et al. (2023). Is my visualization better than yours? Analyzing factors modulating exponential growth bias in graphs. *Frontiers in Psychology*, *14*, http://dx.doi.org/10.3389/fpsyg.2023.1125810.

Menge, D. N. L., MacPherson, A. C., Bytnerowicz, T. A., Quebbeman, A. W., Schwartz, N. B., Taylor, B. N., et al. (2018). Logarithmic scales in ecological data presentation may cause misinterpretation. *Nature, Ecology & Evolution*, *2*(9), 1393–1402. http://dx.doi.org/10.1038/s41559-018-0610-7.

Okan, Y., Janssen, E., Galesic, M., & Waters, E. A. (2019). Using the short graph literacy scale to predict precursors of health behavior change. *Medical Decision Making*, *39*(3), 183–195. http://dx.doi.org/10.1177/0272989X19829728.

Perneger, T., Kevorkian, A., Grenet, T., Gallée, H., & Gayet-Ageron, A. (2020). Alternative graphical displays for the monitoring of epidemic outbreaks, with application to COVID-19 mortality. *BMC Medical Research Methodology*, *20*, 248. http://dx.doi.org/10.1186/s12874-020-01122-8.

Romano, A., Sotis, C., Dominioni, G., & Guidi, S. (2020). The scale of COVID-19 graphs affects understanding, attitudes, and policy preferences. *Health Economics*, *29*(11), 1482–1494. http://dx.doi.org/10.1002/hec.4143.

Schonger, M., & Sele, D. (2020). How to better communicate the exponential growth of infectious diseases. *PLoS One*, *15*(12), Article e0242839. http://dx.doi.org/10.1371/journal.pone.0242839.

Wagenaar, W. A., & Sagaria, S. D. (1975). Misperception of exponential growth. *Perception & Psychophysics*, *18*(6), 416–422. http://dx.doi.org/10.3758/BF03204114.

Wagenaar, W. A., & Timmers, H. (1979). The pond-and-duckweed problem: Three experiments on the misperception of exponential growth. *Acta Psychologia*, *43*, 239–251. http://dx.doi.org/10.1016/0001-6918(79)90028-3.